



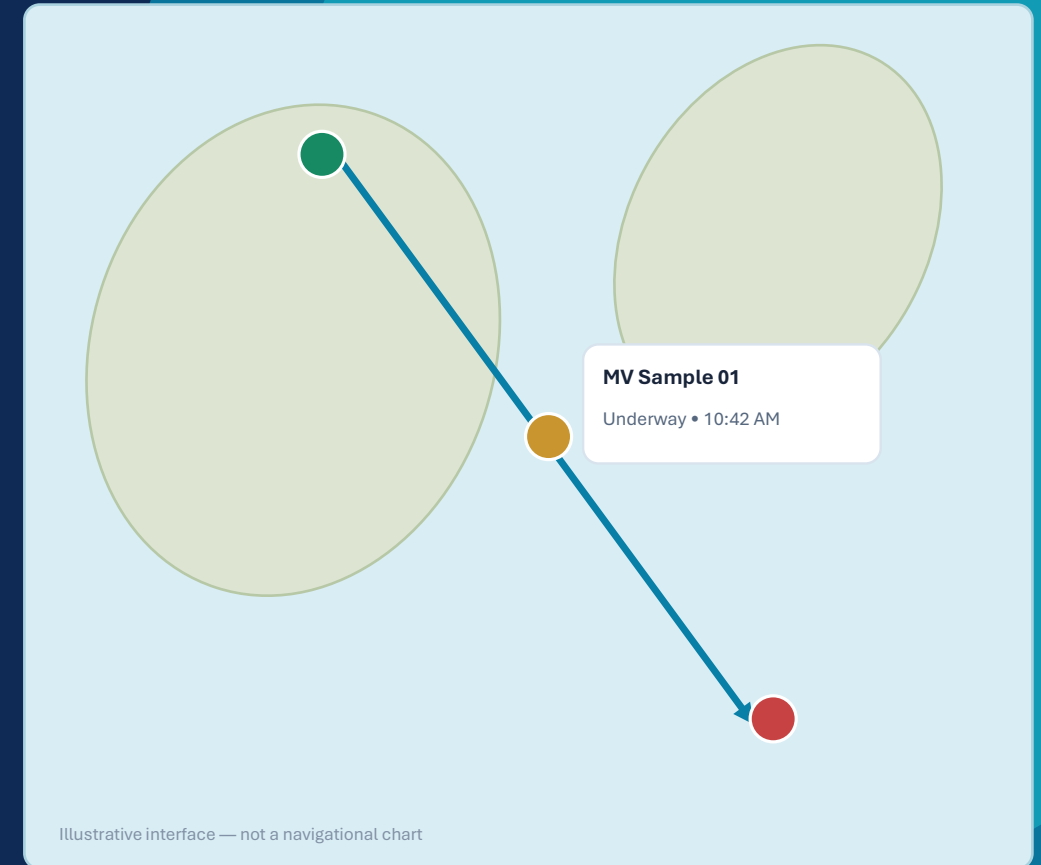
VESSEL MANAGEMENT ENHANCEMENT

Vessel Voyage Map Tracking

A practical, phased proposal for map-based visibility using the existing Voyage and Activity workflow.

CEO PROPOSAL

Prepared for Management Review | 14 September 2026



Villa Shipping Lines, Inc.

One map for the latest known position of every active voyage

This proposal extends the current system; it does not replace the approved vessel and voyage workflow.



Preserve the current process

Vessel → Voyage → Status → Activity → Completion remains unchanged.



Capture coordinates

Origin, destination and activity updates gain latitude and longitude.



Display active vessels

The map shows only vessels with an open/current voyage.

MANAGEMENT OUTCOME

Faster operational awareness without introducing a complex tracking platform on day one.

Managers can open the current-voyage map, identify each vessel’s latest reported position, and open the related voyage details from one screen.

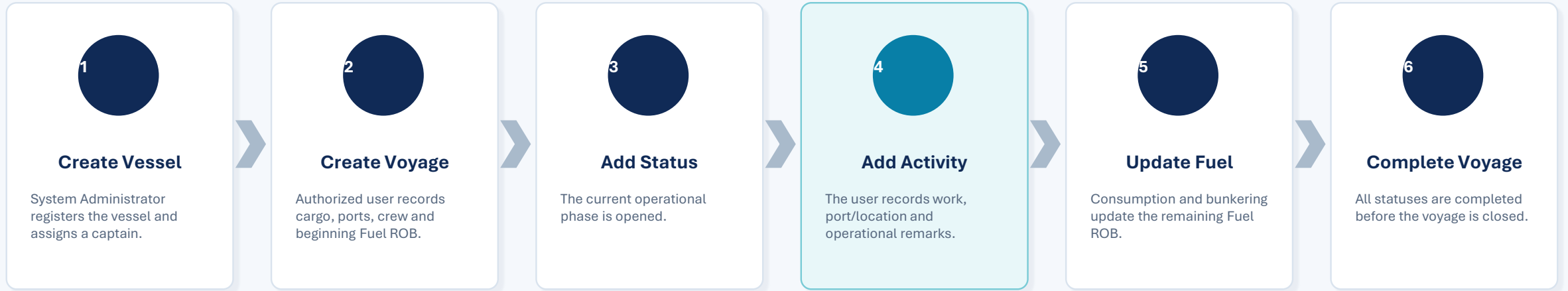
IMPORTANT BOUNDARY

Phase 1 is activity-based location monitoring.

It shows the latest position submitted by an authorized user. Continuous automatic tracking requires a future AIS/GPS integration.

The enhancement connects to the process already in use

No new approval stage is proposed for the initial release.



PROPOSED INSERTION POINT

Coordinates are captured during Create Voyage and each Add Activity update.

The operational workflow stays familiar to vessel personnel.

Current location data exists—but it is not yet spatial

Port and location names are stored today; coordinates and map visualization are not.

CURRENT LIMITATION



Text-only location

Managers must open individual voyage records and interpret location names.



No fleet map

There is no consolidated visual view of vessels with active voyages.



Limited history

Activity locations are stored as text, not as auditable geographic points.

PROPOSED CAPABILITY



At-a-glance visibility

See the latest reported fleet position on one map.



Faster follow-up

Open the current voyage and latest activity directly from a marker.



Foundation for automation

Position history prepares the system for future AIS/GPS integration.

Three coordinated capabilities inside Vessel Management

The design is intentionally simple for vessel personnel and useful for management.

01

Map-enabled voyage setup

- Plot origin
- Plot destination
- Set initial current position
- Retain existing port selections

Voyage origin data

02

Position update in activity

- Use device GPS
- Search a place
- Drop a map pin
- Enter coordinates manually

Operational update point

03

Current-voyage fleet map

- Active voyages only
- Latest position per vessel
- Last update timestamp
- One-click voyage details

Management visibility

Flexible input supports both port and at-sea updates

Every saved position records its source, author and timestamp.

RECOMMENDED USER INTERFACE

Search port or location

Latitude

Longitude

USE MY CURRENT LOCATION

DROP PIN ON MAP

Manual entry remains available when GPS or search is unavailable.



Device GPS

Fastest on mobile. Requires HTTPS and the user’s location permission.



Map search / pin

Best for a known place or a position selected visually.



Manual coordinates

Operational fallback for coordinates obtained from bridge instruments.

Validation: latitude –90 to 90 • longitude –180 to 180 • timestamp required

Current Voyage Map — one marker per active vessel

Only the latest recorded position for an open voyage is displayed by default.

Search vessel or voyage...

5 ACTIVE VOYAGES

LAST REFRESH 10:45

MV Sample 01
Underway • 10:42 AM

SELECTED VESSEL

MV SAMPLE 01

UNDERWAY

Voyage

VL-00048

Latest activity

Departed Cebu

Position

10.3157, 124.8421

Updated

10:42 AM • Capt. Dela Cruz

OPEN VOYAGE DETAILS

Illustrative interface — not a navigational chart

Visibility follows the existing company, position and vessel assignment rules

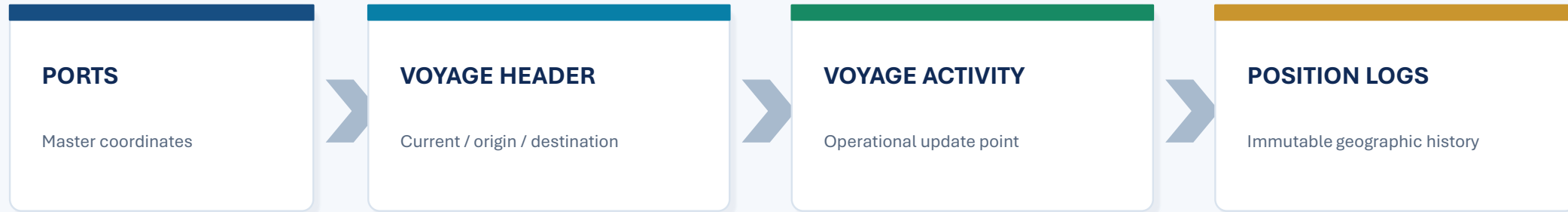
Location features inherit the same controlled access used by Vessel Management and Voyages.

System Administrator	All vessels	Configuration, oversight and troubleshooting
Operations / Vessel Manager	All active voyages	View fleet map and record operational updates
Technical Manager	All authorized vessels	View map and support operational coordination
Vessel Captain	Assigned vessel only	Create voyage and submit current position updates
Executive Viewer	Read-only fleet view	Management visibility without operational editing

Every position update is attributed to a user and timestamp for auditability.

The map is built on current records plus one position-history table

Existing operational data remains intact; new coordinate fields are nullable for legacy records.



NEW COORDINATE DATA

Latitude • Longitude • Accuracy • Source • Recorded by • Recorded at

LEGACY DATA PROTECTION

No existing voyage, activity, fuel or vessel record needs to be deleted or rewritten.

Start practical, then automate when the business case is proven

The initial release establishes usable visibility and the data foundation for later upgrades.

PHASE 1

Activity-based monitoring

Position saved during voyage creation and activity updates.

DATA BEHAVIOR

Human-confirmed updates

RECOMMENDED NOW

PHASE 2

Device-assisted updates

Browser GPS simplifies reporting from an authorized mobile device.

DATA BEHAVIOR

User permission required

OPTIONAL ENHANCEMENT

PHASE 3

AIS / vessel GPS integration

External telemetry automatically updates positions at defined intervals.

DATA BEHAVIOR

Continuous data feed

FUTURE INVESTMENT

Production controls keep the map trustworthy and secure

The map is an operational monitoring tool—not a certified navigation system.



Access control

Company, position and vessel assignment rules determine visibility and editing.



Coordinate validation

Reject invalid latitude/longitude and preserve the selected input source.



Audit trail

Record user, timestamp, activity and previous/current position.



Stale-location warning

Flag markers when no update has been recorded within the agreed threshold.



Mobile fallback

Manual coordinates remain available if GPS or search is unavailable offshore.



Provider protection

Restrict map API keys by domain and keep secrets outside the source code.

Disclaimer: displayed positions are “latest reported” unless and until an automatic AIS/GPS feed is integrated.

A controlled rollout reduces operational and data risk

Each phase can be demonstrated and accepted before the next phase begins.

1

Confirm requirements

Map provider, stale threshold, fields, permissions and final UI approval.

2

Build data foundation

Coordinates, position logs, indexes, validation and production-safe migration.

3

Enhance workflows

Map-enabled Create Voyage and Add Activity screens with mobile support.

4

Deliver fleet map

Active-voyage markers, details panel, vessel filtering and access controls.

5

Pilot and train

Test with selected vessels, validate procedures and prepare user guidance.

6

Production rollout

Deploy, monitor location freshness and collect management feedback.

GO / NO-GO CHECKPOINT AFTER EACH PHASE

Success is measured through visibility, adoption and data freshness

Baseline values should be captured during the pilot before final targets are approved.



Fleet visibility

Active voyages visible on one operational map



Response time

Less time spent locating and contacting a vessel



Accountability

Every position linked to a user, activity and time



Data readiness

Structured history available for future analytics/AIS

SUGGESTED PILOT INDICATORS

 % active voyages with coordinates

 Average age of latest position

 Mobile update success rate

 Map-to-voyage click-through

Approval requested: proceed with Phase 1

Begin with latest-reported visibility and a controlled pilot, while preserving a path to automatic tracking.

APPROVAL SCOPE

- ✓ Approve activity-based vessel position monitoring as Phase 1.
- ✓ Approve a map provider evaluation and controlled API budget.
- ✓ Confirm Operations Managers and Captains as position reporters.
- ✓ Authorize a limited pilot using selected active vessels.
- ✓ Review pilot results before considering AIS/GPS integration.

RECOMMENDED DECISION

PROCEED TO DETAILED DESIGN

Deliverables for the next review:

- Final field list and permissions
- Working map prototype
- Production-safe migration plan
- Pilot acceptance checklist

QUESTIONS & DISCUSSION